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Symbolism and Hesitation: Guideline for Merging Teleoperation with Multiple Users

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Abstract
This paper examines how coordination emerges when multiple users share control of a cybernetic avatar under constrained nonverbal channels. Across three observational studies in stylized testbeds, we identify two recurring regularities: the Maxim of Symbolization, whereby users compress task-relevant intentions into compact, reusable signals, and the Maxim of Hesitation, whereby users avoid direct overrides and often prefer indirect or delayed correction, especially with unfamiliar partners. Rather than claiming ecological representativeness of the tasks, we use these studies to derive design implications for multi-user teleoperation and cybernetic avatar systems. We conclude by specifying the boundary conditions under which these implications are most likely to apply.

Keywords Cooperation work · Symbolism · Hesitation · Guideline

1 Introduction

Thanks to advancements in teleoperation and artificial intelligence technology, it has become possible to perform various tasks remotely or with autonomous assistance that were previously conducted by humans. These advancements enhance human capabilities. For instance, opportunities have been proposed where the elderly or disabled, who traditionally could not actively participate, can now thrive using remote operation and autonomous control technologies [1] Additionally, it has become feasible to control multiple robots simultaneously through teleoperation [2]. Especially under COVID-19, the demand for telecommunication has greatly accelerated [3].

In this context, there is a growing trend to develop technologies that allow multiple people to work collaboratively. For example, its possible for a single individual to control operations in multiple locations simultaneously, or for multiple individuals to simultaneously control two robots, giving the sensation of being in one place [4]. Furthermore, when two individuals control a single robot, actions can be merged seamlessly to provide an integrated experience [5, 6]. The modern notion of extending human physical actions is encapsulated by the concept of the Cybernetic Avatar. The concept of the Cybernetic Avatar (CA) refers to a remotely operated or semi-autonomous robotic system that serves as a proxy for human presence and action. CAs can be controlled by one or more users and are designed to extend human agency across physical and social boundaries, and various technical developments [7] and societal proposals [8] are being conducted for its realization. Furthermore, to realize collaborative control between users and autonomous systems across multiple CAs, new technologies are being developed, such as intent estimation methods using large-scale language models (LLM) and integration with brain-machine interface (BMI) based operations [9].

Numerous attempts have been made, especially in terms of how multiple people can operate a single robot, the control during teleoperation, and the methods of intention mediation. As for the merging of intentions during teleoperation, this has been summarized by Losey and others [10]. Proposed methods of formalization include discretely

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classifying intentions, transitioning intentions continuously, or returning to manual control beyond a certain range.

Most challenging is the mediating mechanism and mediating the high-level intentions of multiple people into a single avatar. High-level intention mediation implies that each individual breaks down the task into multiple subtasks and shares them. Here, high-level intentions refer to goal- or subtask-level representations (e.g., where to place, when to stop, whether to rotate), as distinct from low-level motor commands. By symbolic gestures, we mean compact, repeatable movement patterns whose meaning depends on shared convention rather than literal trajectory alone. For instance, in a task like cooking, one person might cut ingredients while another decorates the dish [11, 12]). In this process, the individuals performing the task must infer each other's intentions and modify their behavior based on each other's movements and actions, such as how they serve and position food items. Therefore, higher-order intention mediation via avatars requires not only the superimposition of multiple people's intentions and their application to robot operations, but also the investigation of how the intentions of multiple people are intentionally mediated via avatars.

Importantly, the baseline assumption that "a single operator can perform the task" is not always appropriate in cybernetic avatar contexts. Operators may have asymmetric capabilities or limited modalities, and effective participation may depend on assistance and role-complementation. Therefore, even when a task is theoretically feasible for a single operator, studying how coordination mechanisms emerge under shared control remains practically motivated.

This paper addresses this gap by shifting the focus from intention mediation between a human and an autonomous system to intention mediation among multiple human operators via a shared avatar and interaction environment. We propose two core principles for designing such systems: the **Maxim of Symbolization**, which refers to the tendency of users to compress and abstract their intentions into symbolic gestures, and the **Maxim of Hesitation**, which highlights the social reluctance to directly negate or override another user's actions. These principles are grounded in observational experiments and supported by linguistic and cultural theories.

This paper makes three contributions. First, it synthesizes observations from three multi-user teleoperation settings into two recurring regularities—the Maxim of Symbolization and the Maxim of Hesitation. Second, it clarifies how these regularities vary across social familiarity and task context. Third, it translates these observations into design-relevant implications for shared-control and cybernetic avatar systems.

This study is an extension of the work of [13], which described unique symbolic and social pressure-induced changes in high-level intention mediation and provided guidelines for teleoperative cooperation principles such as linguistic cooperative Maxim. Specifically, in this paper, we have elaborated on the observational experiments described in the abovementioned research [13] and supplemented them with further observational experiments and the guidelines such observations can provide. However, unlike those prior reports, the present paper integrates the observations across settings and translates them into a unified design framework.

In this study, we focus on the coordination of multiple users in controlling robotic systems. The interactions observed among the users, including non-verbal negotiation and symbolic communication, are inherently social. These social interaction patterns are highly relevant to the field of Social Robotics, where understanding and designing robots that mediate or participate in human social interaction is a central concern.

The structure of this paper is as follows: Sect. 2 outlines the theoretical background and related work. Section 3 explains our observation experiments when users encountered new advanced intention mediations. Section 4 explains the phenomena observed from our experiments. Section 5 explains the guideline design directions derived from the observational findings. Section 6, based on these findings, discusses the guidelines designers must consider when designing multi-user robots intended for high-level intention mediation. Section 7 summarizes these insights, and concludes the study.

2 Background for Tele-Operation Studies with Multiple Users: From Tele-Existence to Cybernetic Avatar

Research in teleoperation has primarily advanced within the field of robotics. Technologically, teleoperation is underpinned by low-latency network technologies, sensing technologies that thoroughly investigate environments, actuator technologies that reproduce precise movements, and haptic technologies that sense and convey tactile feedback. Part of the teleoperation technology is recognized as tele-existence technology, which makes one feel as if they are present remotely [14]. The reproduction of intention in teleoperation has been identified as a challenge [10]. Not limited to the fusion of manipulation of manipulable objects (robots, VR avatars, etc.) to remote locations, which is the subject of this study, researchers are focusing on collaborative work with remote locations via avatars. When using CG avatars, we are dealing with collaborative work in both real and cyberspace, from collaborative work using avatar displays on a display to collaborative work in VR space via mutual

CG avatars. The research deals with collaborative work in both real space and cyberspace, ranging from collaborative work using avatar displays on a display [15] to collaborative work in VR space via each other's CG avatars [16]. Technology that enables robots to work across time and space by operating them remotely is called "remote presence robots," and cooperative work is also being studied in this field. From humanoid robots that interact with humans to wearable robot arms that support the wearer's work regardless of the robot's shape or task, cooperative work using remote presence robots is being studied.

These technologies have extended the experience and capabilities of a single user. The fusion manipulation avatar technology addressed in this study is not limited to extending a single user. However, it is believed that fusion of multiple users' specialties and expertise through avatar manipulation will enable users to go further beyond the limits of their individual abilities without being limited by spatio-temporal constraints.

As an advanced form of this teleoperation technology, the Cybernetic Avatar technology, which involves multiple users in the operation, has been developed [17]. A Cybernetic Avatar offers individuals, who may not be able to actively participate due to physical capabilities or cultural constraints, a proxy body and a degree of anonymity, enhancing opportunities for their active involvement. For instance, even individuals with muscle disabilities who find physical involvement challenging can operate agents by using alternative methods such as tracking eye movements. It has been discovered from related research on the Cybernetic Avatar using a remote-controlled robot named Ory that such individuals can function as cafe staff [18]. The chapter on collaborative control of multiple Cybernetic Avatars in the book compiled by H. Ishiguro [9], which summarizes efforts toward the development of Cybernetic Avatar technology, compiles insights regarding the integration and harmonization of human operation and autonomous control in the operation of multiple Cybernetic Avatars as a direction for the advancement of Cybernetic Avatar technology. Related discussions in smart manufacturing also emphasize real-time feedback, adaptive coordination, and the integration of visual and robotic systems in dynamic human-centered environments [19].

Cybernetic avatar research is pioneering the field of intervention of not only one but multiple operators in avatars, including robots in remote and other locations [20]. When multiple people operate avatars simultaneously instead of a single user, they need to coordinate their operation intentions with each other, and the problems associated with this need to be solved. Hagiwara et al. call such a method of introducing avatars with multiple operators as "Collaborative Avatar" and conducted a verification study [5]. As a result, they verified that the usability is the same as that of a single operator even when two or more operators are working together and that task performance is improved. In addition, Tanaka et al. verified the effect on task efficiency of giving tactile feedback corresponding to the other person's operation in a Collaborative Avatar [6]. The validation results showed that vibration feedback corresponding to speed increased the efficiency of collaborative work. Increasing work efficiency in collaborative operations helps increase the operator's QoL.

Our work aligns with the concept of a "Collaborative Avatar" as proposed by Hagiwara et al., where multiple users jointly operate a single robotic entity. These insights contribute to the broader field of Social Robotics by emphasizing the role of robots as mediators of human-human collaboration, even in simulated or preparatory environments.

In summary, while prior research has advanced the technical feasibility of multi-user teleoperation and intention mediation, our contribution lies in theorizing and empirically validating the social mechanisms—namely, the Maxims of Symbolization and Hesitation—that underpin successful collaboration in such systems. By situating our work within this landscape, we aim to bridge the gap between system-level control strategies and user-level communicative behavior.

3 Observations

Emerging from the evolution of remote operation technology is the Cybernetic Avatar, a transformative innovation enabling multi-human interactions. This technology extends participation opportunities for individuals otherwise restricted by physical or cultural constraints, offering them an alternate body and a level of anonymity. As a testament to its versatility, even individuals with severe muscular disabilities have found a way to control agents through methods such as eye movements [17].

Our previous researches [11, 21, 12], examined how the mediation of each other's intentions occurs in fusion operations, where several people intervene in a single manipulative object. Specifically, we have conducted three observational experiments in two different virtual work environments, as follows.

- Tele-nininnbaori
  - operators who have repeated communication experiences [21]
  - operators do not yet know each other [12]
- Co-topping [11])

These experiments collectively address our research question: how do users mediate intentions in multi-user teleoperation, and how do symbolic and social strategies vary across contexts? By comparing familiar vs. unfamiliar user pairs, and two types of task environments, we were able to identify consistent patterns in symbolic gesture use and hesitation behaviors.

In the following subsections, we briefly describe each experiment, focusing on the aspects most relevant to our theoretical framework. Detailed procedural information is provided in the Appendix.

3.1 Tele-Nininnbaori

Tele-nininnbaori is a manipulation task using nonverbal communication of the same object by multiple persons between remote locations. In Figs. 1 and 2, the remote manipulation object is configured as a CG arm model on the screen. The operators are divided into two groups: the Main-operator and the Sub-operator, each of whom is given the configuration of the operating device. and the authority and task information as shown in Table 1.

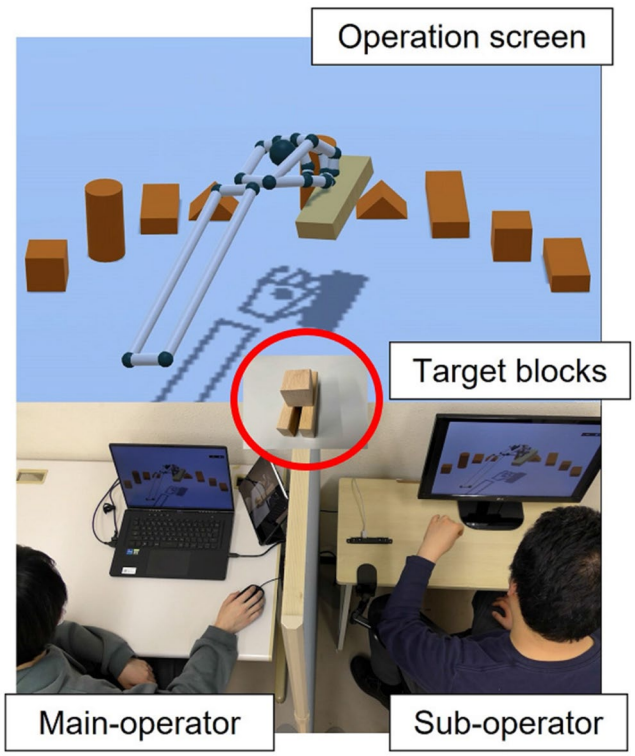


Fig. 1 Multiple-users control: the person on the right is assumed to be the operator, unable to move the arm smoothly, and capable of giving instructions only with a mouse. The person on the left is a sub-operator and can substitute for the movement of the person on the right

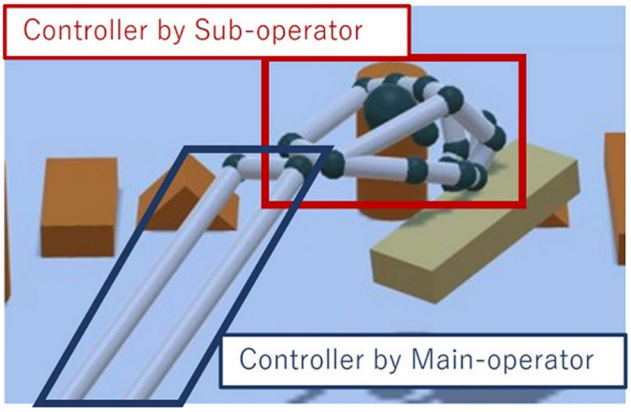


Fig. 2 Separated operation. The model in the figure represents the arms of the avatar on the virtual, with the upper palm controlled by the sub-operator and the lower arm controlled by the operator

Table 1 Division of operators in Tele-nininnbaori

Operator	Operating Device	Operating Authority	Task Information
Main-operator	Mouse	Position of arm	goal of task
Sub-operator	Leap Motion	Hand and finger rotation	nothing

As shown in Table 1, the assignment's goal is assumed to be known only to the Main-operator. In a Tele-nininnbaori task, operators are prohibited from verbal communication. The Main-operator conveys task goals to the Sub-operator solely through nonverbal gestures using a shared manipulation object.

This setup is particularly effective for observing communicative pressure, such as how users handle misunderstandings or express negation. Because gestures are limited to arm movements without fine finger control, misinterpretations are likely, making it easier to observe how users manage disagreement or correction under constrained conditions.

To verify the intention-communication behavior in Tele-nininnbaori, We used the building block task. The Main-operator was the one who controlled the position and knew the goal of the task, and the Sub-operator was the one who owned the flexion of the fingers.

3.1.1 Tele-Nininnbaori with Operators Who Have Repeated Communication Experiences

Our first experiment with Tele-ninnbaori is an observation of the means used to communicate intentions between acquaintances in the relevant environment [21]. This

experiment aimed to investigate how symbolic gestures emerge in collaborative teleoperation when users have prior communication experience. In this experiment, we also conducted trials in which verbal communication was allowed, either unrestricted or partially restricted, for practice.

This setup enabled us to observe how users with shared history develop symbolic gestures—such as directional movements, pauses, and negation signals—to mediate intentions.

3.1.2 Tele-Nininnbaori with Operators Do Not Yet Know Each Other

To investigate the generation of communication pressure in a fusion manipulation in which multiple people intervene in a single manipulation object when the fusing operators do not yet know each other, we conducted an experiment in which two people who had never met each other before performed a fusion manipulation task focused on communication [12]. To examine the generation of communicative pressure in this context, we investigated the strategy of presenting negations or mistakes to the other's actions or perceptions made by one of the manipulators to the other, based on the participants' mentions. We focus on the strategies of negation and pointing out mistakes because negation and mistakes give a powerful message in communication and, therefore, significantly impact specific pressures in communication.

This experiment revealed a notable reduction in explicit negation behaviors compared to the familiar-pair condition. Instead, participants often relied on subtle or delayed responses to signal disagreement. These findings support the Maxim of Hesitation and highlight the influence of social unfamiliarity on communication strategies in shared control environments.

3.2 Co-Topping

3.2.1 System Overview

Co-topping is an experimental system that virtualizes the task of topping pancakes at a remote location using a robotic arm. Figure 3 shows the UI of the system.

Co-topping virtualizes pancake-topping work as a simulation of collaborative avatar-mediated tasks in which two users contribute through complementary roles (e.g., positioning vs. dispensing). We use a virtualized setup to control constraints and observe coordination mechanisms, while maintaining a clear link to plausible real-world collaborative topping contexts.

Moving and placing a model of the robot arm on the virtual pancake shown in Fig. 3, a circle (CreamDot) that imitates cream can be placed on the pancake to simulate toppings, such as drawing some design on the pancake. Assuming that the experimental system is limited in operation and that few operations are possible, all system operations can be performed only by moving and left-clicking the mouse. Dragging operations are not supported.

The brown circle in the center square area in Fig. 3 is the pancake model, and the gray rectangle with blue triangles is the robot arm's paw model. The yellow and green spheres follow each operator's mouse cursor and indicate the mouse cursor's position on the other side.

The operator can move the robot arm model to the clicked position by clicking within the central square area. The position can be specified even during the movement. If the position is determined during the movement, the model moves to the re-specified position linearly from the current position.

The UI on the left side of Fig. 3 is for adjusting the model placement in the pancake topping, which simulates the application of cream. The operator can control the start and

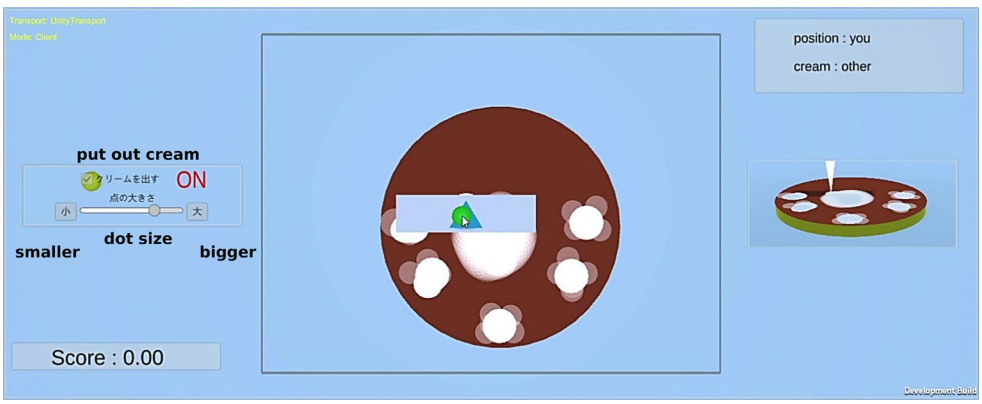


Fig. 3 UI of use system

stop of CreamDot placement by the checkbox at the top. The left and right buttons on the bottom slide bar control the size of the CreamDot to be placed and how often they are placed. The size and frequency are increased or decreased simultaneously. This specification reflects the situation in which, when topping cream on an actual robot, the operator can only adjust the amount of pressure by manipulating the finger part.

The display in the upper right corner of Fig. 3 shows the status of the operator's responsibilities. The operator is responsible for moving the robot arm (Position) or adjusting the model placement with the cream (Cream) application. On the right side of each job description, "you" or "other" is indicated, and the job marked with "you" is the one the operator is currently in charge of. In the UI shown in Fig. 3, the operator is in the role of Position. In the UI of the co-operator, "you" and "other" are interchanged, indicating that the co-operator is playing the role of Cream.

The display on the right side of Fig. 3 is an image from another viewpoint camera of the pancake model. From the center viewpoint camera, it is impossible to see the details of the placement of the cream model because of the superimposed arm model. Therefore, the image is presented from a view of the pancake directly beside it. The white elongated triangle extending from the top in the picture is the tip of the cream squeezer model attached to the arm. From a different viewpoint, the video is taken from the side of what is possible in implementing the actual device.

The Score displayed in the lower-left corner of Fig. 3 shows the participants' reproduction accuracy at a specific point in time for a particular pattern. As described in detail below, it is assumed that the experimental task is to reproduce a particular design by operating the UI, and the degree of reproduction is presented to the experimental participants as a factor in judging the progress of the task and its completion. For the degree of reproduction, the similarity between the task and the execution result is calculated using the similarity judgment based on Accelerated-KAZE (AKAZE) feature detection implemented in OpenCV.

Apart from Fig. 3, an administrative UI shares the same work situation. The administrative UI does not have operating privileges on the robot arm model but has meta-operating rights in the work environment. Specifically, it can replace operation privileges, score reproducibility, and initialize the environment by deleting all cream dots. The work results are also saved in the administrative UI.

3.2.2 Observation Experiment

In this observation study, we tasked the participants with reproducing a specific pattern presented to them by Co-topping system. The interface allowed only mouse-based input, and verbal communication was not restricted but was rarely used, especially among unfamiliar pairs.

Despite the simplicity of the interface, participants developed symbolic strategies for coordination, such as using cursor motion to indicate shape, timing, or emphasis. In particular, we observed that even unfamiliar pairs were able to achieve successful collaboration through non-verbal cues alone.

This experiment complements the previous two by demonstrating that symbolic mediation strategies can emerge across different task types and interaction modalities. It reinforces the general applicability of the Maxims of Symbolization and Hesitation in multi-user teleoperation.

4 Findings

We present the findings on intention mediation obtained in each observational experiment described in Sect. 3. The results are organized around two recurring patterns observed across all experiments: the emergence of symbolic communication and the influence of social dynamics on how users express disagreement or correction.

4.1 Emergence of Symbolic Communication

Across all experiments, participants developed and refined non-verbal gestures to mediate intentions. These gestures evolved from direct, shape-based actions (e.g., dragging to indicate position), to movement-based cues (e.g., rotation gestures), and eventually to more abstract, symbolic signals (e.g., horizontal shaking to indicate negation).

In the Tele-nininnbaori task with familiar operator pairs, symbolic gestures were used more frequently and with greater confidence. Participants developed shared conventions, such as using vertical movements to indicate confirmation and horizontal movements to signal disagreement. These conventions were reinforced through repeated interaction and mutual adaptation.

In contrast, unfamiliar operator pairs used fewer symbolic gestures and relied more on literal or movement-based cues. The absence of shared context appeared to delay the emergence of abstract representations, although some symbolic behaviors still emerged over time.

In the Co-topping task, symbolic communication was also observed, particularly in the use of cursor motion to indicate shape, timing, or emphasis. Even when verbal communication was permitted, many participants preferred non-verbal strategies, suggesting that symbolic mediation is not only effective but also intuitive in shared control environments.

Representative interaction patterns included (i) transitions from shape-based indications to movement-based cues

and then to conventionalized symbolic signals, and (ii) indirect correction strategies such as pausing, withholding the next action, or repeating a request instead of explicit negation. Additional procedural and stimulus-level details are retained in Appendix A for readability.

4.2 Indirect Communication and Social Sensitivity

Participants' strategies for expressing disagreement or correction varied significantly depending on their familiarity with their partner. In unfamiliar pairs, users rarely employed direct gestures to indicate errors. Instead, they relied on indirect strategies such as pausing, repeating actions, or waiting for the other user to self-correct. These behaviors reflect a sensitivity to social dynamics and a preference for preserving harmony in unfamiliar interactions.

In contrast, familiar operator pairs were more likely to use explicit gestures to guide or correct their partners. This suggests that social familiarity reduces the perceived cost of direct intervention, enabling more efficient coordination.

In the Co-topping task, similar patterns were observed. Unfamiliar pairs often avoided verbal correction, even when allowed, and instead relied on subtle cursor movements or timing adjustments. These findings reinforce the idea that social pressure influences not only what users communicate, but how they choose to communicate it.

4.3 Summary

The findings from all three experiments highlight two important aspects of multi-user teleoperation: the natural development of symbolic communication as a means of coordinating action, and the role of social context in shaping how users' express disagreement or correction. These insights provide a foundation for designing systems that support both effective and socially sensitive collaboration.

Taken together, these observations suggest that successful multi-user teleoperation depends not only on combining control inputs but also on supporting the emergence of shared symbolic practices and socially sensitive forms of correction.

5 Design Guidelines

Through these experiments in Chapter 3, 4, we have gained insight into higher-order intention mediation in cybernetic avatars and designed guidelines. We describe them in the following sections.

5.1 From Shape to Symbolization

A situation in which both parties do not know the context of the other is the most difficult situation for intention mediation. Imagine a situation where two types of extraterrestrials are trying to communicate. Initially, they won't understand each other and will have to mimic actions to guess the intention. If you are intelligent, you would likely recognize incorrect actions by the other party and either correct them or wait for them to realize their error. Bates' study indicates that symbolization in language is becoming more common [22]. Our previous research [21] showed a transition in human-to-human communication from mimicry to signaling. Here are three gestures.

Shape Gestures: These gestures predominantly involve using a mouse to directly indicate a specific shape. For instance, when determining the position of a triangle from a top-down view, operators often draw it out in advance. Additionally, instructions to align the longer side of a block vertically are frequently given through vertical mouse movements.

Movement Gestures: These primarily encompass rotation gestures, where the direction of rotation is indicated by mimicking the desired arm movement. Operators use the mouse to guide the object to a particular location and then demonstrate the rotation. While this method is intuitive, especially for those unfamiliar with intention mediation, it demands more from the operator. During intention mediation, co-operators actively use the mouse to guide whether a negative movement should occur. Besides rotation, a frequent instruction was to drop a block, indicated by a top-to-bottom mouse motion.

Symbolic Gestures: The most prevalent symbolic gesture was a horizontal waving motion signifying negation. Rotation presents a notable challenge in block assembly. While position can be conveyed via arm movement, orientation is trickier with just mouse movement. Thus, a prevalent strategy was to use a negative gesture. The process generally involved the operator moving a selected block to a specific spot. Once positioned, the co-operator would automatically rotate it. If the rotation was incorrect, the operator signaled a negation. Horizontal gestures, often used negatively in human communication, were similarly employed in non-verbal contexts. Interestingly, symbolic gestures familiar in face-to-face interactions were also prominent in Tele-Nininbaori.

In summary, communication gestures evolved through three stages:

Shape Gestures Directly drawing or indicating shapes (e.g., tracing a triangle).
Movement Gestures Mimicking actions like rotation or dropping.
Symbolic Gestures Abstract signals such as horizontal shaking to indicate negation.

This progression reflects increasing abstraction and shared understanding between users (see Fig. 4).

This progression—from shape-based gestures to movement-based cues and ultimately to symbolic gestures—can be interpreted as a form of social learning and meaning-making. This aligns with the theory of Symbolic Interactionism proposed by George Herbert Mead [23], which posits that individuals construct shared meanings through interaction, particularly through the use of gestures and symbols. We found that this evolution of gestures was influenced by several factors:

Time and repeated interaction In familiar operator pairs, symbolic gestures such as horizontal shaking to indicate negation emerged more quickly and were used more consistently. This suggests that shared history and repeated exposure facilitate the development of abstract communicative conventions.
Practice and task familiarity Even in unfamiliar pairs, symbolic gestures began to appear in later stages of the task, particularly after participants had completed practice trials. This indicates that symbolic communication can emerge through task-based learning, even in the absence of prior social familiarity.
Task complexity More complex tasks, such as those requiring precise coordination of rotation and placement, appeared to accelerate the shift from literal to symbolic gestures. In these cases, symbolic gestures offered a more efficient and less ambiguous means of communication.

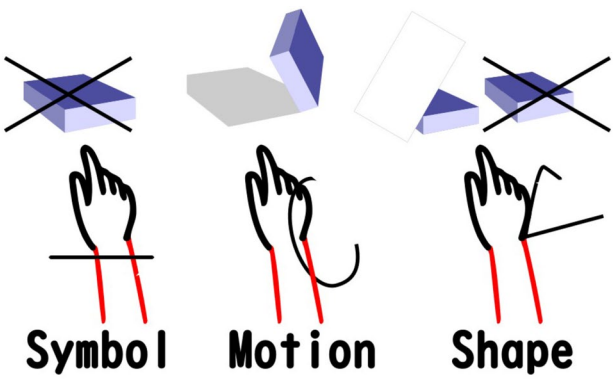


Fig. 4 Examples of gesture types observed in teleoperation: (**a**) Shape gestures, (**b**) Movement gestures, (**c**) Symbolic gestures

These findings suggest that symbolic communication is not static but develops dynamically through interaction. As shown in Fig. 4, participants moved from direct shape indication (e.g., drawing a triangle with the cursor) to movement-based cues (e.g., rotating the arm to suggest orientation), and finally to symbolic gestures (e.g., horizontal shaking to indicate "no").

This progression reflects not only cognitive adaptation but also social negotiation. Participants were not merely signaling intent—they were co-constructing a shared language of action. This insight supports the idea that teleoperation systems should be designed to support the emergence and refinement of such symbolic practices over time.

This suggests that the principles of language also apply to body movements. Humans can interpret minor vertical and horizontal gestures as affirmative or negative, guiding their actions and intentions [24]. However, our studies indicate that while humans freely use denial gestures in standard remote operation situations, they alter their communication strategy in constrained scenarios like a fixed neck, prompting others to mimic them [25].

When both parties share the context of the other, the above stages and beyond are strongly influenced by the nature of the task and the operating environment. In another previous study, participants in an observational experiment used both the shape and movement gestures described above and did not use symbolic gestures. This phenomenon was also in pairs who had extremely little verbal communication. The nature of the task first influenced this phenomenon. As the task used was figure drawing, the emphasis was on the movement trajectory of the manipulated object to draw and the sharing of the assumed drawing result. The operating environment also had an impact. Although participants limited the operations to mouse movement and clicking, the shared gestures of shape and trajectory were used for intentionality mediation, as the participants could share each other's operation trajectories. In short, as mutual predictability increases, participants shift from depicting task geometry (shape gestures) to depicting action dynamics (movement gestures) and finally to using conventionalized symbolic signals that efficiently encode confirmation or rejection.

5.2 Social Pressure: Hesitation for Denial

If the above principle of symbolization were the only principle, then intention mediation in gesture could in principle be achieved by directly affirming or denying the other's action. However, there is dignity among cooperating humans, and direct negation can be an attack on the dignity of the other party. Therefore, the operator needs to mediate intentions in light of these social pressures.

This behavior aligns with the principle of indirectness in pragmatics [26]. In many East Asian cultures, for example, indirectness is a preferred communication strategy to avoid confrontation and reduce social tension. This aligns with the findings of Pang et al. [27], who showed that Chinese participants were more adept at interpreting indirect nonverbal cues than British participants and were more likely to use such cues to convey disagreement. These findings suggest that the cost of direct intervention is not only cognitive but also social.

Several studies assume that when multiple people control a device, they will act optimally to achieve their goal. But if the psychological barrier of directly denying a partner's action outweighs the benefit of achieving a task, instructions can become more complex. Especially in tasks that require high-level intention mediation, these issues become evident. Our research found that when tasks are divided between arm and hand movements, the person controlling the arm usually takes the lead. However, those controlling the hand actions do not actively deny the arm's movements. Instead, they engage in sabotage by not following up with the subsequent required action, indirectly signaling the primary operator to recognize their error [12].

5.3 Pragmatic Parallels in Language: Avatar Manipulation is Another "Language" But Nonverbal

The principles of symbolization in physical instructions and social pressures show similarities with pragmatics in linguistics [28]. Pragmatics is the study of how language is used in actual communication. Human communication isn't just about exchanging dictionary-defined words, but also about the context in which they're used. The word "flatten", for example, can mean to roll out dough or to clean up a room, depending on the context. Similarly, gestures, like using the palm of the hand to level a surface, can also convey intention.

Our previous research found that when two people remotely assemble an object, they not only operate the robot but also use its movements to convey intentions [21]. In this task, two individuals were responsible for different functions: one controlled the arm movements and the other the hand movements of a single robot to complete a block assembly challenge. During this task, it was observed that users did not simply manipulate the robot. Instead, they used the robot's motions to communicate their intentions. For instance, the user in charge of controlling the arm deliberately moved it vertically or horizontally to convey affirmation or negation, as illustrated in Fig. 5.

These actions are not random; they follow specific patterns. Communication happened through simple affirmative or negative signals or by tracing shapes the other party focuses on. This aligns with the cooperative principle identified by Grice in linguistics. Grice noted that ideal communication involves providing information that's both quantitatively and qualitatively comprehensible to the listener [29]. Our research suggests that this principle is vital even in seemingly non-communicative tasks like remote operations. As more remote operations become automated in the future, this cooperative Maxim remains essential. It's a well-known principle in human-AI collaboration, and AI systems adhering to this Maxim outperform those that don't [30].

This linguistic consideration is also related to the principle of "respect" explained in the next section.

5.4 Similarity of Respect in Language: Intervention for Intention is Considered Rude

We also discovered that the evolutionary principle of politeness, as proposed in pragmatics, is present during instructions as well. Generally, in any language, it's undesirable to negate someone else's intentions [31]. For example, directly demanding actions from someone (using directives like "Please") is considered rude because it directly intervenes

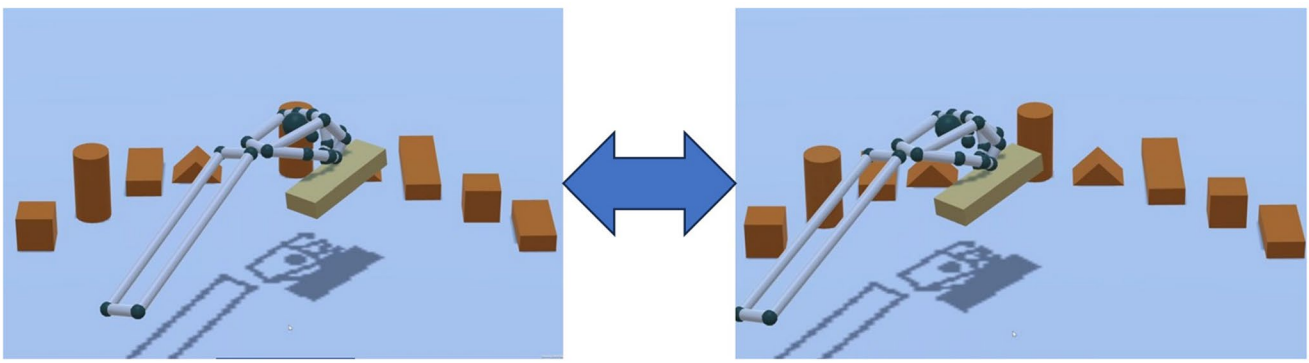


Fig. 5 Denial signal: the user conveys a negative response by swinging the arm. This gesture serves as a non-verbal indication of disagreement or refusal

with the other person's intention. In these situations, it is more preferable to ask about the other person's capability (using terms like "Could") or to inquire if they have the necessary experience (using terms like "Have"). Based on our past research, we found that this same principle applies to remote operations as well. In our experiments, strangers interacted with each other. Unlike previous studies, we didn't observe any actions that directly negated the other person's actions. Instead, when a user noticed a mistake in the other person's action, they simply chose not to proceed with the next action. This allowed the other person to realize their mistake on their own. This behavior can be seen as an application of the principle of politeness to remote operations, suggesting that intervening in someone else's intentions is considered rude.

While these regularities resonate with classic pragmatics and symbolic interactionism, our contribution lies in showing how they materialize in shared-control teleoperation, where communication and action share the same channel and interventions have immediate operational consequences. This setting raises design-relevant coordination problems that are not directly captured by conversation-only accounts.

6 Guidelines for Tele-Operation with Multiple Users

In light of the above, we believe that the following principles should be considered when designing teleoperation controls that mediate higher-order intentions. Of course, the degree to which these considerations are desirable depends on the situation. For example, in emergency situations such as disaster response, military action, or medical practice, it may not be desirable to consider the mediation of such roundabout intentions. Even in such situations, however, problems may arise, for example, when the rescuer feels hesitant toward the rescued. The guilt-ridden victim may not affirm the rescuer's instructions and wait for the instructions to be changed, which is detrimental to the overall benefit. On the other hand, in situations where more routine teleoperated robot capabilities are expected, such as cooking, light work, or assisting the elderly or disabled, the principles of symbolism and sensitivity to social pressure are of fundamental importance.

The insights gained from our simulation-based experiments may inform the design of physical robotic systems. While the robotic arm used in our study was virtual, the interaction patterns observed—such as indirect negation, symbolic gestures, and role-based coordination—mirror those expected in real-world collaborative robot use. This reinforces the relevance of our proposed guidelines to Social Robotics, where robots are expected to support or mediate human-human interaction in socially appropriate ways.

6.1 Maxim of Symbolization

If users must coordinate through a shared actuator under limited nonverbal channels, the system should provide compact and repeatable signaling resources (e.g., small symbolic motion vocabularies or low-cost intent cues), because users tend to compress their intentions into abstract, reusable forms. Therefore, it's crucial for system designers to always provide mechanisms that can compress user intentions. For instance, having dedicated signals for negation can aid in intention mediation among users. For example, in one of the Cybernetic Avatar studies, disabled users collaborated with each other on a pancake-painting task[1]. Here, the user is supposed to apply cream to a pancake and arrange fruit on top of it according to the user's request at a café. At this point, since there are several patterns of how to apply the cream, we believe that the operators can abstract its signals according to the situation to facilitate intention mediation.

Additionally, learning from users' actions and having the system assist based on those learned intentions can facilitate smooth operational integration among users. Such automation also functions when using artificial intelligence technologies to assist user actions.

6.2 Maxim of Hesitation

If operators are unfamiliar, role-asymmetric, or socially cautious about direct intervention, the system should support mediated correction and gradual action transitions rather than relying only on explicit overrides, because disagreement may be expressed indirectly or with delay. For instance, when two users who meet for the first time mediate intentions, they would opt for a strategy that indirectly allows the other party to realize something, rather than explicitly negating it. In relative communication with a person whom one has never met before, one may hesitate to explicitly deny the other's action, resulting in a far-fetched negative expression. In the remote negotiation task of this study, i.e., the joint manipulation of a single object, only the participant knows the goal of the task and can be considered to be in control of the task. This social pressure in communication may have caused inhibition or reticence in behavior. In the case of avatars operated by multiple participants, the goal of the task is to control communication.

Social pressures of communication may significantly inhibit certain types of interaction, depending on the nature

[1] Collaborative Avatar experiment in AVATAR ROBOT CAFE DAWN ver. β, https://cybernetic-being.org/activities/202211_collaborative_avatar_cafe/, (Accessed: 2023–09-01.).

of the task. In situations where all operators share the goal of the task, these social pressures may limit the expression of more explicit verbal interactions to intention mediation about the task's difficulty. In this case, operators tend to make most intention mediation consist of signals on the shared environment.

System designers should be mindful of social pressures among users and ensure that alternative means are available that do not increase users' social pressure. For example, a system design could be such that when a user's behavior is not as intended, the system, not the user, indirectly corrects the behavior to make it clear that no other user interference took place. This would reduce the psychological load on the user. In this case, however, it is necessary to make the fused operators fully aware of the background beforehand to avoid a deceptive form of interface, which could be criticized in persuasion engineering [32]. Another approach is to make it explicit that both operators share the goal of the task and that they have the initiative in the operation. Methods such as modifying behavior without revealing it to users can facilitate this consideration.

6.3 Design Implications for Shared-Control Systems

The two maxims identified in this study can be treated as design constraints for shared-control systems in which a robot or avatar mediates coordination between multiple users. Rather than adding a separate communication channel, such systems should be designed with the assumption that users will use the shared action channel itself to negotiate, signal, and adjust intentions.

Under the Maxim of Symbolization, systems should provide compact and repeatable signaling resources that allow users to express task-relevant intentions in stable, reusable forms. In practice, this may include a small vocabulary of symbolic motion elements—such as pause, cancel, handoff, or rotation cues—that can be expressed through short trajectory primitives, posture changes, or other low-cost movement patterns. The goal is not to prescribe a universal language, but to support the emergence and consistent use of symbolic forms that users can quickly appropriate in context.

Under the Maxim of Hesitation, systems should not assume that explicit override is always the most appropriate mode of coordination. Especially when users are unfamiliar with one another or occupy asymmetric roles, disagreement may be communicated indirectly through pauses, delayed responses, or subtle corrective actions rather than direct contradiction. Design implications therefore include mediated correction, gradual action transitions, and low-cost mechanisms that allow users to signal mismatch or resistance without requiring abrupt interruption.

These implications also suggest a possible role for AI-assisted shared control. In such settings, symbolic and hesitation-related cues may serve as interpretable markers for detecting intention shifts, anticipating coordination breakdowns, or mediating transitions between user actions and autonomous assistance. Such cues include low-amplitude preparatory movements that can be used as early warnings before larger actions. We clarify that such low-cost signaling mechanisms are not introduced ad hoc: they are derived from observed preparatory or low-amplitude movement patterns that preceded larger task-relevant actions and functioned as early intent announcements. We emphasize, however, that this is a design translation from the present observational results rather than a direct validation of a specific AI feature.

6.4 Scope and Boundary Conditions

Our studies employ stylized constraints to elicit coordination mechanisms under (i) a shared actuator where action and signaling share a channel, (ii) limited nonverbal bandwidth, and (iii) possible role/knowledge asymmetry across operators. We do not claim that the specific tasks are ecologically representative of all teleoperation settings; rather, we claim that the observed regularities inform the design of systems where such constraints and asymmetries are expected. Validating downstream design features in fully realistic physical deployments remains future work.

7 Conclusion

(8–2)In this study, we examined how coordination mechanisms emerge when multiple users share control of a single avatar under constrained nonverbal channels. Across three observational studies, we identified two recurring regularities: the Maxim of Symbolization, whereby users compress task-relevant intentions into compact and reusable signals, and the Maxim of Hesitation, whereby users often avoid direct overrides and instead rely on indirect or delayed correction, especially in unfamiliar pairings. These observations suggest that multi-user teleoperation may be designed not only as a control problem but also as a socially mediated interaction environment. These observations inform design-relevant implications for shared-control systems, as discussed in Sect. 6, but their

effectiveness in physical robot implementations remains to be tested.

Rather than claiming that our tasks are ecologically representative of all teleoperation settings, we use them as stylized testbeds to derive design-relevant implications for systems in which operators share a single actuator under limited communicative bandwidth. In such settings, compact signaling resources, mediated correction, and socially cautious action transitions may improve coordination.

(8–6,9–4)Future work should examine how these implications transfer to more realistic physical deployments, how they vary across cultural and relational contexts, and how AI-assisted shared-control systems can detect and interpret symbolic and hesitation-related cues without overstepping users' intentions.

Appendix A Details of Our Previous Research Experiment

A.1 Tele-Nininnbaori with Operators Who Have Repeated Communication Experiences

A.1.1 Method

Three participants (A, B, and C) alternately took charge of the Main-operator and Sub-operator, and tested both cases of communication in language and cases of communication only in non-language. Even when communicating in language, only the operator speaks, and the sub-operator does not. As a result, a total of 12 cases have been implemented. The operator and accompanying operator are taught how to operate in advance, and they try it well by themselves, and work after getting used to the interface. The time is a total of 10 minutes and failure is considered if not completed within 10 minutes. The building blocks given as the target of the main task, used in non-verbal, consisted of four blocks and three levels in height as shown in Fig. A1.

The data collected were the task execution time and intention-communication behavior during execution. The latter was the central observational data collected by observing the behavior during the experiment and by the mentions of the experimental participants in the post-event interviews.

A.3.2 Results and Findings

Among the experimental results, we analyze these cases because we are particularly interested in intention mediation in non-verbal situations. We have summarized below what signals have been seen in specific cases of non-verbal situations (Fig. A1).

Case 3: The target of the building block task was easy to imagine, and the operation succeeded without the inquiry from the sub-operator only by the operator carrying the arm. Finally, the sub-operator expressed success by making a good sign with his thumb up.

Case 4: The work was not completed. Non-verbal communication from the operator to the accompanying operator is to suggest a movement to approach by moving up and down, to convey negation by falling sideways, and to convey a different direction by moving quickly to the right. There was a movement that showed a change of direction by shaking it up.

Case 7: The work was not completed. For the operator or accompanying operator, there were actions such as showing negation by moving sideways, adjusting the position, and presenting the direction. There was an action which indicated the correct direction by moving it up and down. In addition, there was a movement to indicate the direction of rotation by turning. In addition, by moving it into a triangle, a movement indicating the recognition of a triangular prism when viewed from above was observed. We could also observe the act of giving an advance notice of the movement by lifting and dropping.

Case 8: There were 8 cases in which the operator or sub-operator denied the work by shaking it sideways. In addition, the movement to transmit the grasping release was observed by the downward movement.

Case 11: There were 3 cases in which the operator or sub-operator denied the work by shaking it sideways. In addition, by performing the rotational motion, there were four movements that presented the rotational direction.

Case 12: The work was not completed. There were 7 cases in which the operator or accompanying operator denied the work by shaking it sideways. In addition, by performing the rotational motion, there were 8 movements that presented the rotational direction. There was also movement from top to bottom to show the direction.

A.2 Tele-Nininnbaori with Operators Do Not Yet Know Each Other

A.2.1 Method

The experiment participant participated in the task as the Main-operator and the Sub-operator was one person on the experimenter's side for all trials. We set up three tasks: one practice task and two main tasks, one of low difficulty and the other of high difficulty. The practice task was the same for some participants, but the main task differed for all

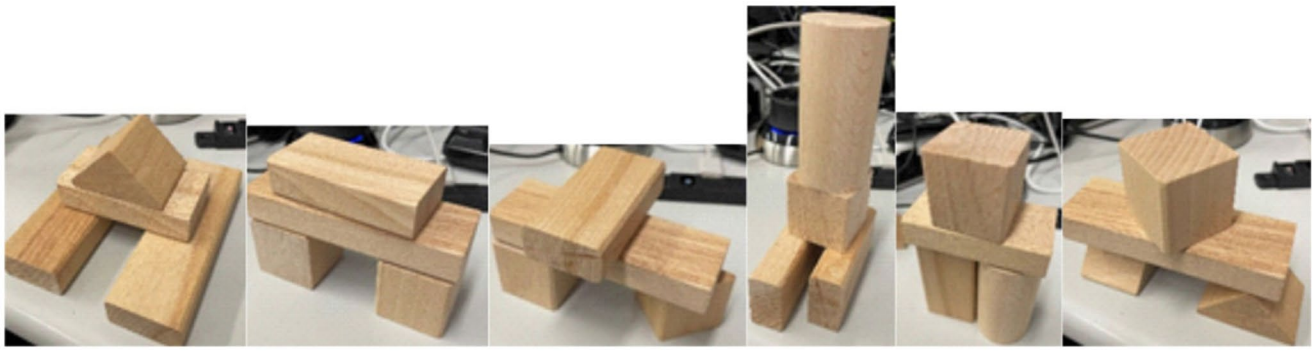


Fig. A1 Among the building blocks, those used in non-verbal communication. From left, case 3, case 4, case 7, case 8, case 11, case 12. Each of them consists of four blocks and three layers

participants in all trials. Details of the difficulty settings are shown in Table A1. The number of rotations required is the number of times the operators must rotate the building block before achieving the goal. The number of times to put them on is the number of times another block is placed on top of it. We counted the number of times to put them on by the number of times the placing action was performed, regardless of the number of building blocks below it.

Examples of the various tasks are shown in Fig. A2 below.

The order in which we presented the tasks was always low difficulty followed by high difficulty. This order was because, although practice tasks were performed, participants who were operating the Tele-nininnbaori system for the first time on the day of the experiment were made to complete the high-difficulty tasks before they had much experience, which could result in them performing complex tasks without being able to formulate a policy for communicating their intentions, which might make it difficult for them to perform the communicative behavior.

We recruited 21 participants. We excluded one participant from the data because they misunderstood the description of the experiment and performed the gesture with his arm instead of the arm on the screen. The 20 participants who were not excluded were ten males and ten females each, ranging in age from 19 to 49 years (ave.=26.2, st.dev = 8.81). The experimenter in charge of the Sub-operator confirmed that they had never met all participants. We paid all participants, including those who were excluded from the data, an experimental fee of 1,650 yen.

The experimental procedure was shown in below.

1. Description of the experiment
2. Informed consent
3. Explanation of operation using the actual system
4. Implementation of the exercise
5. 1 minute break
6. (4–5 again with different tasks according to participants' requests)
7. Execution of the low-difficulty task of the main experiment
8. Answers to questionnaire
9. 5-minute break
10. Execution of the high difficulty task of the main experiment
11. Answers to questionnaire
12. Conducting interviews

The maximum duration for the practice task was 5 minutes, and the maximum period for the main task was 10 minutes. Only for the main task, when the whole time limit was reached, the task was terminated at that point. The participant's judgment determined the completion of the task, and we provided the following criteria for the determination.



Fig. A2 Blocks the difficulty of each block was designed according to Table A1

Table A1 Block difficulties

Difficulty	Building blocks	Turns required	Times to put them on
Practice	2	0∼1	1
Low difficulty	3	0∼1	1
High difficulty	4	1 or more	2

- The type of building blocks used must be consistent.
- The positional relationship and arrangement of the building blocks were consistent (within one-third of the length of the sides of the cube in the execution environment).
- The orientation of the building blocks, including the entire structure, should be consistent (within a margin of error of 30 degrees or less in terms of angle).

When the participants completed the experiment, the supervisor pointed out only if the type of building block used was incorrect and encouraged them to correct it. We included the time for pointing out and fixing in the execution time.

The participants and the Sub-operator were introduced to each other only during the explanation of the experiment, and no chatting was conducted to build a relationship between the participants and the Sub-operator. In addition, the participants and the Sub-operator were not allowed to exchange signatures during all experiment phases, from the investigation explanation to the practice session and breaks. The participants were allowed to speak only during the practice task, given that they were unfamiliar with the operation itself. Still, only some talked to the experimenter in charge of the Sub-operator.

Experimental data were collected through questionnaires and interviews.

The questionnaire obtained the following three items on a 7-point Likert scale.

- Sense of Agency: I felt that I was moving the CG arm.
- Sense of Body Ownership: I felt as if the CG arm was my own arm.
- Task Satisfaction: I am pleased with the quality of the building blocks we have assembled.

In the interviews, a semi-structured format was used with the following content as a starting point.

- How much was there an overall sense of self-manipulation of the hand being worked on?
  - If there was a difference between the first and second sessions, what factors contributed to the difference?
- What was the percentage of contribution to the work?
  - Were there any differences between the first and second sessions?
- Intentional Mediation Behavior
  - What intentions and behaviors did you try to communicate?
    * Did you perform any behavior that indicated a behavioral error or miscommunication? If so, what did you do?
  - What was the success rate of communication?
    * Was there a difference between the first and second attempts?
  - Were there any instances of anticipation, where the Sub-operator moved before the communicating action?
- What was your impression of the Sub-operator?
  - Were there any changes between before and after the start of the experiment?
- Other
  - Other points of interest
  - Impressions of the experiment
  - Questions for the experimenter

Among the data collected, the central analysis item is the behavioral strategies that indicate negation in the intention-to-mediate behavior questioned in the interviews (Did you perform any behavior that indicated a behavioral error or miscommunication? If so, what did you do?). By analyzing the behavioral strategies that indicate denial, we will examine the occurrence of communication-related pressure.

A.2.2 Results and Findings

Classifications were made for the behavioral negation strategy by the experiment planner and those who participated in the experiment as assistant operators. Table A2 shows the results of the classification and assignment of the denial behavior strategies mentioned by the participants; the participant in ID 1 was excluded from the table because we didn't individually interview them about their denial behavior.

The classification items were created based on a generic view of the negative behavior strategies taken by the

Table A Classification of subjective comments on negative behavioral strategies

Classification	Behavior examples	Experimenter allocation	Sub-operator allocation
Active Presentation	Signs using the metaphor of negation	Wave the hand		
	Repeat change orders	Start over	2,4,10,12,13, 14,15,16,17,18	4,10,12,13,14,15,16,18
		Continue the approach		
		Repeat rotational orders		
	Special change orders other than above	Move the hand in the direction they want it to go back	19	9,19
		Use different building blocks to tell the correct orientation		
Passive presentation	Wait for sub-operator to change	Not moving to the next step in the process	3,9,20	2,3
		Not moving to the assembly location		
		Do nothing		
Not presentation	Couldn't think of any	-	6,8,11	6,8,11,20
	No occasion to point out mistakes	-	5,7	5,7,17

participants, in addition to the "hand waving" strategy, which was the most frequently reported negative behavior strategy in previous studies. The assignment of the adverse action strategies the participants took to the classification was based on the list of negative action strategies collected through the interviews. When there were descriptions that were difficult to classify, the classifiers listened to the audio data of the relevant interviews as supplementary material to understand the exact nuances of the responses and made more accurate classifications.

- No duplicate classifications were allowed. If more than one item was considered applicable, priority was given in the following order of the major categories
  - Active Presentation > Passive Presentation > Not Presentation

    * the order of priority is set by the order in which the participant's sense of ownership of the action is considered high.

- If you think that more than one item falls under one major category, classify them into the one that the participant considered more important in the interview voice
- Some participants mentioned their strategies and those they thought about taking but did not. In such cases, regardless of the priority of the major category, classify based on the strategy taken
- The number of IDs divided among the classifications may differ significantly. We do not assign IDs that are unlikely to fall into fewer classifications. It is acceptable to have no IDs assigned to some classifications

We calculated Krippendorff's α [33] as the goodness of fit between the two classification results, which is 0.708. From the books [34] and papers [35] of the proponents, 0.667 is stated as a lower limit of the number that can be considered a tentative conclusion of the agreement. Therefore, this classification is consistent to some degree.

Each of the classification items and the discussion are described in detail below.

Active Presentation: Signs using the metaphor of negation This classification strategy is based on using the entire arm of the CG as a negation behavior, mimicking the behavior of swinging the head and hands sideways, which is a metaphor for No in general. In the paper that proposed the Tele-nininnbaori task [21], this behavior was reported to be the most common in communication. However, in the present experiment, we classified the participants' mentions of the negation behavior strategy and found that no applicable cases were present in the 19 participants of the valid data set.

Active Presentation: Repeat change orders This classification was based on the strategy in which the participant indicated a denial by repeating an instruction when the other party's current state or action was incorrect.

Although there were some differences in the assignments made by the two groups, this classification was considered applicable by the most significant number of participants.

Active Presentation: Special change instructions other than above This classification is based on the strategy that participants indicated their rejection of the status quo by taking particular actions that did not fall into the two categories listed above. As outlined in Table 1, specific examples of this strategy include directional instructions to return to the previous state or indicate the direction in which the current state should be changed using another building block.

Passive Presentation: Wait for sub-operator to change

This classification is a strategy to show the negation of the current state by not proceeding from one step to the next in the building block assembly process. In addition to simply doing nothing, other strategies include not moving to the placement operation or not going to the assembly location after separating the assembly location and the orientation adjustment location.

Not Presentation: Couldn't think of any

This classification is the case in which the participants did not mention the negation behavior strategy. This case includes the circumstances in which no behavior was performed as a negative behavior in the experimental trial or the communicating behavior itself was hardly performed. In some cases, the behavior was similar to the negative presentation strategy described above, but the participants' behavioral intentions differed.

Not Presentation: No occasion to point out mistakes

This classification is the one in which the participant mentioned that the situation in which the negation should have been made did not occur in the first place. In this case, as far as the participant remembered, all of their communicative actions were received correctly, so there was no need to implement a denial strategy. However, some of the participants who were classified stated that the behavior they were considering as an adverse action strategy did exist.

The classification results show that participants took none of the strategies revealed in previous studies to indicate to the supporter the difference between the current situation and the direction of rotation by waving the hand sideways. This strategy is although this is the most frequently cited essential communication in previous studies. This difference may be attributed to social pressure in communication. When communicating relative to a person one has never met before, explicitly negating the other person's behavior is sometimes hesitated and becomes a far-fetched negative expression. In the tele-negotiation task in this study, i.e., the joint manipulation of a single object, only the participant knows the goal of the task and can be considered to be in control of the task; it is thought that this social pressure in communication caused the inhibition of action or reticence. In the case of avatars operated by multiple people, the goal of the task is to control the communication. The fact that the occurrence of refraining was observed in avatars used by multiple people means that the communication necessary for the execution of the task may need to be smoother among the participants. This problem should be considered when introducing the multiple-operator avatar technology to society.

A.3 Co-Topping

A.3.1 Method

In the experiment, Twenty participants were recruited. The details are described in section 3.4, but participants were to work in pairs for this verification. Therefore, we allowed single and paired applicants to participate in the experiment. In the case of paired applicants, the pair that applied was considered to participate in the experiment. In the case of a single applicant, he/she was paired with someone who had similarly used alone to participate in the experiment. Of the ten pairs, eight were pairs who requested to participate in the experiment as a pair at the time of application, and two were pairs that combined applicants who applied alone. Both pairs that combined loan applicants had never met before at the time of the experiment. The gender of the participants was ten men and ten women. The age range was 19 45 (ave.=22.95, st.dev = 6.30).

In this validation, we tasked the participants with reproducing a specific pattern presented to them by Co-topping system. All the tasks given in this verification are shown in Fig. A3.

The tasks had three difficulty levels: practice, low, and high. There are two types of tasks for each difficulty level since a different task is presented at the time of role change. All participants were expected to make all assignments. As for the order of presentation, the practice task was given in a typical order (T→X) for all pairs, but we presented these tasks in a different order for each pair to remove the order effect. In this case, to avoid giving a high-difficulty task at a low level of skill even though the participants were practicing, we continually presented the first two sessions of this task in a low-difficulty order, and we gave the latter two

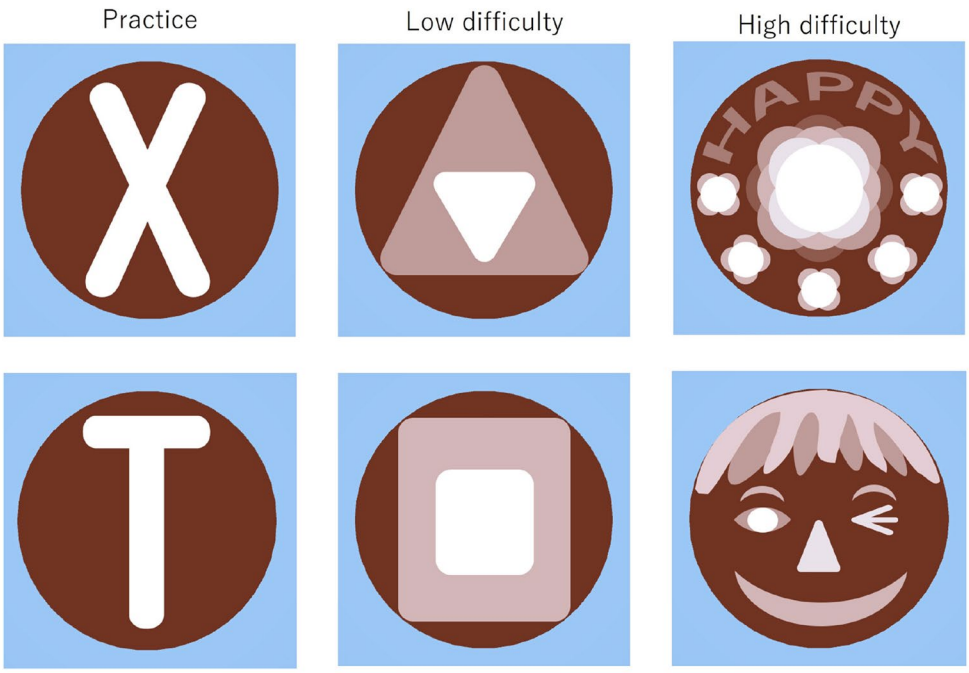


Fig. A3 All tasks

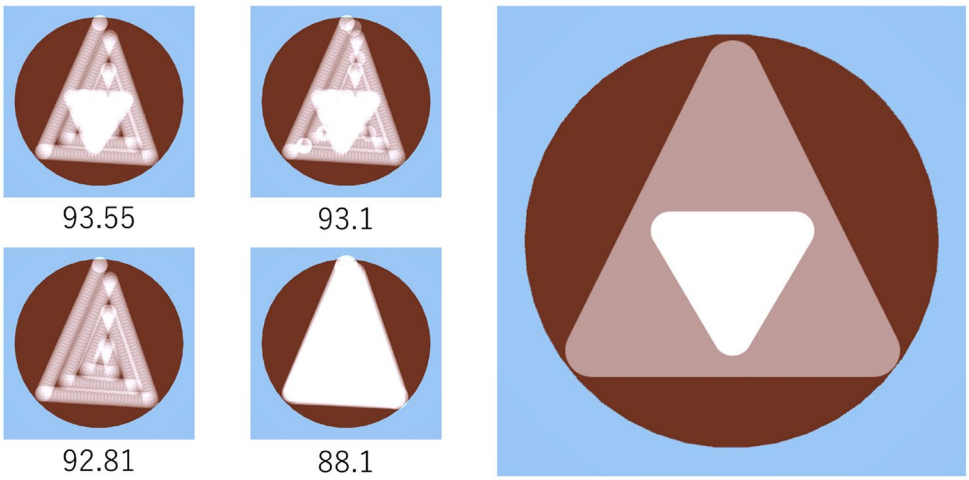


Fig. A4 Task showing example

sessions in a high-difficulty order. We changed the order by selecting one of the tasks that complied with this condition. The differentiation of task difficulty is based on the complexity of the shape to be realized and the number of color shades used. In other words, practice and low-difficulty tasks have simple shapes, while high-difficulty tasks have complex shapes. In addition, the intensity of the colors used increases as the experiment progresses from the practice, low-difficulty, and high-difficulty tasks. In the high-difficulty task, the figure has a tapered shape and other features that can be reproduced more accurately by adjusting the output by the cream staff while the robot arm model is moving. When presenting the tasks, we also gave four examples of each of the reproducibility of the experimenter's implementation, along with the task itself, as shown in Fig. A4.

The execution of the task by the experimenter was performed by the system used, which was adjusted so that a single person could operate it.

The experimental procedure is as follows.

1. Description of the experiment
2. Informed consent
3. Explanation of the operation using the actual system
4. Practice exercise
5. 1-minute break
6. Repeat 4–5 with different tasks and operators
7. Perform the low-difficulty task of the main experiment
8. Answering the questionnaire
9. 5-minute break
10. 7–9 again with different tasks and operators

11. Perform the high difficulty task of the main experiment
12. Answering the questionnaire
13. 5-minute break
14. 11–12 again with different tasks and operating personnel
15. Conduct the hearing

The practice task was conducted with an upper limit of 5 minutes per task, while the main task was conducted with an upper limit of 10 minutes per task. Only in the main assignment, the assignment was terminated when the maximum time limit was reached. Completion of the task within the time limit was subject to the agreement of the two participants, and as a criterion for judgement, the task was presented with the condition that it should be completed with a higher reproducibility by referring to the reproducibility of the execution examples accompanying the presented task, as shown in Fig. A4.

In addition, during the explanation of the operation, the following operations were presented as possible by request to the experimenter by agreement of the participant pairs.

- Initialising the environment
- Scoring the reproducibility
- Completion of task implementation (final scoring)

In the event that time ran out, the reproducibility scoring at the end of the task was also presented to the participants.

In the explanations and hearings, the participants were separated by partitions and two participants were supposed to receive the explanations and answer the hearings at the same time, two at a time. During the operation of the system, the participants worked back-to-back across the partition, looking at each other's screens.

After explaining the experiment and giving informed consent to the two participants, an explanation of the operation of the actual system was given to the two participants who were to be paired with. They then performed the practice task twice, switching roles each time. The main task was then performed twice in the low-difficulty task and twice in the high-difficulty task, with the roles changing after each session. After the main task, the participants were asked to complete a questionnaire each time, and two simultaneous interviews were conducted at the end of the task. There was also a break between each assignment.

Experimental data were collected through questionnaires and interviews. The questionnaire items and scales were as follows.

- Contribution
  - Please select the number that best describes the percentage of this topping work that you feel you were involved in (X/10) [11-point scale of 0∼10].
- Task satisfaction
  - Please select the most applicable statement about the performance of the topping task. [7-point Likert scale]

The interviews were conducted in a semi-structured format, using the following content as a starting point. In addition, as two people were to be interviewed at the same time, each person was asked to take turns in referring to the other.

- What was your contribution to the overall four tasks? (Refer to questionnaire responses if necessary).
- What was the level of satisfaction with the execution of the whole four tasks? (Refer to questionnaire results if necessary)
- Apart from conversations, were there any other means used to communicate your intentions to the other person or to capture their intentions?
- Other
  - Other points of interest
  - Comments on conducting the experiment
  - Questions for the implementer

A judgement is also made on the image of the result of the task as to whether it was successful or not. The judgement is made by observing the images of the result of the work to determine whether the elements determined for each level of difficulty are met. The specific elements for each difficulty level are as follows.

- low difficulty
  - Is the shape of the general frame correct?
  - Is the difference in density between the large frame shape and the inner shape expressed?
- high difficulty
  - Are the parts that make up the pattern complete?
  - Are the shapes of the parts generally consistent?
  - Is the arrangement of the parts generally consistent?
  - Are the differences in shading between the parts expressed?

In the high difficulty task, success was considered if some elements were met, even if not all elements were fully met.

Among the data collected, the central items for analysis are the items on judging the success of the images of the work product and the means used in the interviews to communicate intentions to the other person and to capture the other person's intentions, other than through conversation.

Results and Findings

We judged a success on the results of 40 tasks performed by the participant pairs for ten tasks with four times per pair. First, in the low-difficulty task, all three factors were satisfied all 20 times. For the high-difficulty task, few of the work results could express the shading of each part. However, the remaining three elements were generally satisfied in all trials, although some still needed to be included. Therefore, we considered that the participants succeeded to some extent in all the trials, and we will analyze the interviews from all pairs.

References in the interviews to means of communicating intentions other than conversation included, for almost all pairs, the presentation with the mouse of content about the position of the cream application and the trajectory of the line. In a particular case, one pair indicated the thickness and density of the cream by specifying it against a control bar of the size of the cream.

Outside of the interviews, we observed an extreme lack of verbal communication in pairs who had never met each other before. This phenomenon means that these pairs could mediate their intentions only through non-verbal communication using the mouse pointer on the screen. As mentioned above, the fact that these pairs also succeeded in executing the task suggests that the intention mediation was successful.

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Data Availability The datasets generated during and/or analysed during the current study are not publicly available due to the fact that the original data is voice data containing the naked voices of the experiment participants but are available from the corresponding author on reasonable request.

Declarations

Ethical Approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards. This study was conducted in accordance with the Academic Integrity Code of Keio University (Approval number: 2022–113, 2022–128).

Informed Consent Informed consent was obtained from all individual participants included in the study.

Competing Interests The authors have no relevant financial or non-financial interests to disclose.

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